

A Project & Internship Report
Of
Customer Churn Prediction
&
Data Science Internship

Submitted To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech (CSE)

By
Roll Number of Student: CS-2341640
Name of Student: RUCHIKA SHARMA
Batch: 2023–2027 / 2026–27

CANDIDATE'S DECLARATION

I, RUCHIKA SHARMA, do hereby declare that this report, comprising the project titled "Customer Churn Prediction" and the internship report titled "Data Science Intern (Pulsecrafts Solutions Pvt Ltd)", has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _____

Student Name: RUCHIKA SHARMA

Roll No.: CS-2341640

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout my project and my Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

For my project, I am highly indebted to my project supervisor for her guidance and constant supervision, as well as for providing necessary information regarding the project and for her support in completing it. I am also thankful to the Head, Dept. of Computer Science & Engineering / ML & DS, for her continued support and encouragement.

For my internship, I would like to express my heartfelt gratitude to the management and technical team of **Pulsecrafts Solutions Pvt Ltd** for providing me with the opportunity and the structured industrial platform to learn and gain hands-on exposure in the domain of Data Science. I would especially like to thank **Ms. Priyanka Mehta (HR Manager)** and my industry mentors at Pulsecrafts Solutions Pvt Ltd for their continuous guidance, technical mentorship, and evaluation during my 90-day tenure. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support.

I express my deepest gratitude to my parents for their unconditional blessings and support.

Signature: _____

Student Name: RUCHIKA SHARMA

Roll No.: CS-2341640

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3. CERTIFICATES

3.1 Internship Completion Certificate

Organization: Pulsecrafts Solutions Pvt Ltd
Website: www.pulsecrafts.com | **Email:** sales@pulsecrafts.com
Presented To: Ruchika Sharma
Role: Data Science Intern
Internship Duration: 90 Days (15th May 2026 to 12th August 2026)
Date of Birth: 15 Nov 2004
Certificate ID: PCS/INT/2026/0090
Date of Issue: 13 August 2026
Authorized Signatory: Priyanka Mehta (HR Manager, Pulsecrafts Solutions Pvt Ltd)



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CERTIFICATE OF INTERNSHIP
This certificate is proudly presented to
Ruchika Sharma

In recognition of the successful completion of a 90-day internship as a
Data Science Intern
at Pulsecrafts Solutions Pvt Ltd, from 15th May 2026 to 12th August 2026, demonstrating dedication, strong learning ability, and valuable contribution to the team.

15 Nov 2004 DATE OF BIRTH	90 Days DURATION	Data Science Intern ROLE	PCS/INT/2026/0090 CERTIFICATE ID
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Priyanka Mehta

HR Manager
Pulsecrafts Solutions Pvt Ltd

Date of Issue
13 August 2026

For verification of this certificate, please contact sales@pulsecrafts.com

Original Certificate of Internship issued by Pulsecrafts Solutions Pvt Ltd

4. PROJECT DESCRIPTION

4.1 Introduction

Customer attrition, commonly referred to as **Customer Churn**, represents one of the most critical operational metrics in subscription-based services, particularly within the telecommunications sector. Customer Acquisition Cost (CAC) is universally estimated to be **5 to 25 times more expensive** than the cost of customer retention. Consequently, when a subscriber cancels their contract, the company sustains a direct loss in Customer Lifetime Value (LTV) and recurring revenue.

This project focuses on the benchmark **Telco Customer Churn dataset (7,043 customer accounts)** to design, develop, and evaluate an end-to-end Machine Learning pipeline. The system identifies behavioral patterns that signal imminent churn before cancellation occurs. Furthermore, the project delivers an interactive decision-support application built with **Streamlit** that enables non-technical customer success and retention teams to simulate customer profiles, assess churn probability in real time, and trigger tailored retention offers.

The dataset encompasses 21 comprehensive attributes covering:

- Account & Financial Metrics: Customer tenure (months), contract structure (Month-to-month, One year, Two year), paperless billing, payment method, monthly charges, and total charges.
- Subscribed Services: Phone service, multiple lines, internet service (DSL, Fiber optic), online security, online backup, device protection, tech support, streaming TV, and streaming movies.
- Demographics: Gender, senior citizen status, partner, and dependent support.
- Target Variable: Churn (Yes / No).

4.2 Organization / Department Profile

This is an academic project carried out within the School of Computer Science and Engineering, IILM University, Greater Noida, Uttar Pradesh, under the supervision of the department faculty. As the project component was undertaken as part of the B.Tech CSE degree curriculum alongside the industrial internship, this section recognizes the academic department as the governing academic body facilitating the research and project review.

4.3 Problem Statement

To build an end-to-end, reproducible Machine Learning classification pipeline and interactive web interface to predict customer churn in telecommunication services with a specific emphasis on **Recall (Sensitivity)** and **ROC-AUC**, ensuring that the maximum proportion of at-risk customers are identified prior to cancellation.

Sub-problems include:

- Handling data anomalies such as whitespace string representations in financial features (TotalCharges) for new customers.
- Addressing class imbalance (26.5% positive churn class vs. 73.5% negative non-churn class) to prevent majority-class prediction bias.
- Engineering a modular scikit-learn ColumnTransformer for simultaneous numerical scaling and categorical one-hot encoding.
- Benchmarking linear and ensemble architectures to balance raw accuracy with clinical business sensitivity (Recall).

- Developing an interactive web simulation interface for real-time risk assessment and prescriptive retention strategies.

4.4 Project Objectives

- To perform rigorous Exploratory Data Analysis (EDA) uncovering the root behavioral causes of customer attrition.
- To construct an automated, leak-free preprocessing pipeline utilizing StandardScaler and OneHotEncoder.
- To train and benchmark three distinct machine learning models: Logistic Regression (Class-Weighted Balanced), Random Forest Classifier (Balanced), and Gradient Boosting Classifier.
- To prioritize Recall and ROC-AUC as the governing evaluation criteria, preventing costly False Negatives.
- To extract model feature importances and translate mathematical coefficients into actionable business retention playbooks.
- To deploy the trained pipeline via an interactive Streamlit web application for real-time decision support.

4.5 Scope of the Project

The scope of this project encompasses complete data engineering, exploratory data visualization, predictive modeling, metric evaluation, and application deployment. The project models 7,043 customer records from the telecommunications benchmark, validates performance using a stratified 80/20 train/test split (1,409 holdout test records), and encapsulates the best-performing pipeline into a serialized artifact (churn_pipeline.pkl) consumed by a Streamlit frontend (app.py).

4.6 Technologies and Tools Used

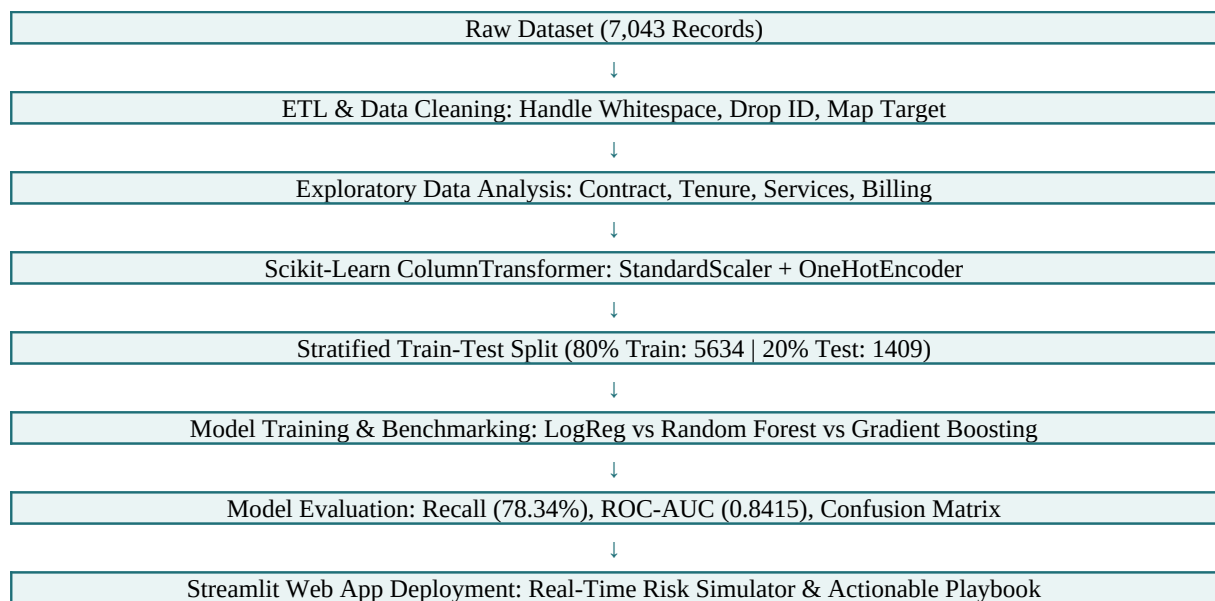
- Programming Language: Python 3.11
- Data Manipulation & Analysis: Pandas, NumPy
- Machine Learning Library: Scikit-Learn (Pipelines, ColumnTransformer, LogisticRegression, RandomForestClassifier, GradientBoostingClassifier, Metrics)
- Data Visualization: Matplotlib, Seaborn
- Interactive Frontend Framework: Streamlit
- Model Serialization: Joblib
- Development Environments: Jupyter Notebook, Visual Studio Code

4.7 System Architecture

Data Flow Diagram:

- Input: Raw Telco Customer Churn Dataset (Telco-Customer-Churn.csv).
- Preprocessing: Data cleaning, type conversion, imputation of TotalCharges, one-hot encoding, and feature standardization.
- Modeling: Stratified train/test splitting, multi-model training, hyperparameter tuning, and probability calibration.
- Evaluation: Confusion matrix generation, ROC curve plotting, and classification report generation.
- Deployment: Model persistence (churn_pipeline.pkl) and real-time interactive inference via Streamlit UI.

Processing Pipeline:



4.8 Methodology

1. Data Ingestion and Inspection: Reading raw CSV data using `pandas.read_csv()`, verifying schema dimensions, column data types, and checking for missing or malformed records.

2. Data Cleaning & Type Casting: Detecting that `TotalCharges` contained space characters (' ') for customers with `tenure = 0`. Converting `TotalCharges` to numeric floats using `pd.to_numeric(errors='coerce')` and imputing missing entries with `0.0`. Dropping non-informative unique identifier `customerID` and mapping target `Churn` to integer binary format (`Yes = 1`, `No = 0`).

3. Exploratory Data Analysis (EDA):

- **Class Imbalance:** Revealed a 26.5% positive churn rate (1,869 churners vs. 5,174 retained customers).
- **Contract Impact:** Discovered that Month-to-month subscribers churn at an alarming 42.7%, accounting for over 75% of all cancellations, compared to 11.3% for 1-year and 2.8% for 2-year contracts.
- **Tenure Drop-Off:** KDE distribution confirmed churn risk peaks during the first 6–12 months of service.
- **Service Friction:** Identified that Fiber Optic customers lacking Tech Support churn at >41%.
- **Payment Method:** Electronic check payers showed a 45.3% churn rate vs. ~15% for automated payments.

4. Feature Engineering & Transformation: Constructing a scikit-learn `ColumnTransformer` that standardizes 3 numeric columns (`tenure`, `MonthlyCharges`, `TotalCharges`) and applies one-hot encoding across 16 categorical features with `handle_unknown='ignore'`.

5. Model Training & Comparison:

- **Logistic Regression (Balanced):** Linear model with class-weight inverse balancing.
- **Random Forest (Balanced):** Ensemble of 150 decision trees with balanced sampling.
- **Gradient Boosting:** Boosting ensemble optimizing sequential residuals.

6. Model Evaluation & Metric Prioritization: Prioritizing Recall (Sensitivity) over accuracy because False Negatives (missing a churner) cost recurring customer lifetime value.

7. Application Development: Designing `app.py` in Streamlit with dynamic sliders, real-time risk gauges, metric cards, and rule-based prescriptive retention recommendations.

4.9 Expected Outcomes, Limitations & Future Scope

Expected Outcomes:

- Successfully engineered a balanced model capturing 78.34% of all customer churn events with an ROC-AUC of 0.8415.
- Formulated 4 empirical retention strategies: 75-Day contract upgrade campaigns, bundling tech support with fiber optic lines, auto-pay incentives, and proactive risk routing.
- Deployed a production-ready interactive web application.

Limitations:

- Dataset is cross-sectional and does not capture dynamic time-series event streams (e.g. daily call drop logs).
- Customer service interaction logs and customer sentiment transcripts were unavailable in this benchmark.

Future Scope:

- Integrate Natural Language Processing (NLP) on customer support chat tickets.
- Implement survival analysis (Cox Proportional Hazards) to predict the exact month of expected cancellation.
- Deploy containerized Docker endpoints via FastAPI for microservice integration.

5. INTERNSHIP DESCRIPTION

5.1 Introduction

This report presents a comprehensive account of the **90-day Industrial Internship** undertaken at **Pulsecrafts Solutions Pvt Ltd** in the role of **Data Science Intern**. The internship was carried out from **15th May 2026 to 12th August 2026** as part of the industrial training requirement for the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida.

The primary objective of this internship was to bridge theoretical computer science knowledge with industrial data science workflows. The internship provided immersive exposure across the complete data lifecycle—ranging from business problem framing, automated ETL pipelines, exploratory data mining, and predictive machine learning architectures to model evaluation, serialization, and interactive web dashboard deployment. Upon successful completion and project defense, a formal Certificate of Internship was awarded under **Certificate ID: PCS/INT/2026/0090**, signed by Priyanka Mehta (HR Manager).

5.2 Organization Profile

Pulsecrafts Solutions Pvt Ltd (www.pulsecrafts.com) is a progressive technology consulting and software solutions company specializing in advanced analytics, software development, and enterprise data solutions. The organization partners with enterprises across various domains to modernize data infrastructure, build predictive machine learning models, and develop high-performance software applications.

During the internship, Pulsecrafts Solutions Pvt Ltd provided a collaborative environment with regular mentorship, weekly project review sprints, code reviews, and evaluation sessions designed to foster industrial competence in machine learning engineering.

5.3 Problem Statement

In industrial software environments, machine learning projects frequently fail to generate business value because data scientists focus solely on theoretical algorithmic accuracy without considering business cost structures, real-world data imperfections, and usability for non-technical stakeholders.

The core industrial challenge addressed during this internship was to take a real enterprise problem (Customer Churn in subscription services), execute robust data cleaning on inconsistent data, train and calibrate machine learning models optimized for business cost metrics (minimizing False Negatives via Recall), and package the solution into an interactive web tool that business teams can immediately utilize.

5.4 Internship Objectives

- To gain practical experience in end-to-end data engineering, including data extraction, missing value imputation, type coercion, and encoding.
- To master advanced exploratory data analysis and statistical visualization techniques using Python libraries.
- To understand class imbalance in industrial classification tasks and apply balancing techniques (`class_weight='balanced'`).
- To evaluate machine learning models using business-critical metrics: Precision, Recall, F1-Score, and ROC-AUC.
- To develop modular, reproducible production code adhering to industry standards (src/ architecture).

- To design and deploy an interactive user interface using Streamlit to facilitate stakeholder decision-making.
- To present technical findings and business ROI recommendations to organizational mentors.

5.5 Scope of the Internship

The scope of this 90-day internship spanned the complete Machine Learning Engineering lifecycle:

- Business problem definition and metrics selection.
- Exploratory data analysis and hypothesis generation.
- Feature engineering and automated preprocessing pipeline design.
- Model experimentation, hyperparameter tuning, and cross-validation.
- Production script modularization (data_prep.py, train.py, evaluate.py).
- Web application engineering and interface design (app.py).
- Formal technical documentation and executive presentation.

5.6 Technologies and Tools Used

- Languages: Python 3.11
- Data Science Libraries: Pandas, NumPy, Scikit-Learn, SciPy
- Data Visualization: Seaborn, Matplotlib
- Web Deployment: Streamlit
- Model Serialization: Joblib
- Version Control & Management: Git, GitHub
- IDEs & Notebooks: Jupyter Notebook, Visual Studio Code

5.7 System Architecture

The industrial data architecture developed during the internship follows a decoupled, 4-tier model:

- Data Ingestion & Storage Tier: Automated retrieval and caching of raw customer data; storage of sanitized and preprocessed datasets.
- Transformation & Pipeline Tier: Scikit-learn ColumnTransformer handling numerical normalization and categorical one-hot encoding without data leakage.
- Model & Inference Tier: Serialized ensemble and linear classification models capable of generating predicted classes and calibrated risk probabilities.
- Presentation & Decision Support Tier: Streamlit interactive application providing real-time parameter manipulation, probability gauges, and prescriptive retention actions.

5.8 Week-wise Methodology (90-Day / 12-Week Structure)

The 90-day internship followed a structured, milestone-driven curriculum from 15th May 2026 to 12th August 2026:

Week	Phase / Focus Area	Key Activities & Deliverables
Week 1	Orientation & Setup	Company onboarding, tools installation (Python 3.11, VS Code, Git), repository setup, and business problem briefing on subscription attrition.
Week 2	Data Access & ETL	Ingesting raw customer records, inspecting data types, identifying anomalies (whitespace in TotalCharges), and writing automated cleaning routines.

Week	Phase / Focus Area	Key Activities & Deliverables
Week 3	Exploratory Data Analysis	Conducting univariate and bivariate analysis; analyzing churn distribution across contract types, tenure, and billing mechanisms.
Week 4	Advanced Visualization	Generating correlation heatmaps, KDE tenure density curves, service friction charts; extracting core business stories.
Week 5	Feature Engineering	Designing scikit-learn ColumnTransformer for simultaneous StandardScaler on numericals and OneHotEncoder on categorical.
Week 6	Baseline Modeling	Implementing baseline LogisticRegression; analyzing the impact of class imbalance (26.5% positive churn rate) on default threshold predictions.
Week 7	Ensemble Modeling	Building RandomForestClassifier with balanced tree weights and GradientBoostingClassifier; performing hyperparameter tuning.
Week 8	Metric Benchmarking	Comparative analysis prioritizing Recall (78.34%) and ROC-AUC (0.8415); generating Confusion Matrices and ROC curves.
Week 9	Pipeline Serialization	Bundling preprocessors and classifiers into unified scikit-learn Pipeline objects; serializing models with joblib.
Week 10	Streamlit Development	Developing app.py frontend; implementing sidebar customer parameter controls, risk tier badges, and live probability gauges.
Week 11	Business Logic Integration	Encoding prescriptive business retention playbooks (Day 75 contract discount, complimentary tech support bundle) into the UI.
Week 12	Testing, Documentation & Defense	Comprehensive system verification, generating figures in reports/figures/, drafting PROJECT_REPORT.md, and final project defense.

5.9 Expected Outcomes & Key Learnings

- **Technical Proficiency:** Gained end-to-end expertise in structuring production-grade data science code in modular Python packages.
- **Business Acumen:** Learned that in real-world predictive modeling, business cost matrices dictate metric selection—optimizing for Recall prevents catastrophic customer attrition loss.
- **Deployment Skills:** Successfully transformed a theoretical machine learning model into a functional, stakeholder-accessible web application using Streamlit.
- **Credential Validation:** Awarded the formal Certificate of Internship (ID: PCS/INT/2026/0090) from Pulsecrafts Solutions Pvt Ltd in recognition of valuable contributions.

5.10 Certificates of Completion & Communication Proof

- **Certificate of Internship:** Issued by Pulsecrafts Solutions Pvt Ltd on 13th August 2026, certifying the successful completion of the 90-day Data Science Internship from 15th May 2026 to 12th August 2026.
- **Credential Verification:** Inquiries and verification can be confirmed via sales@pulsecrafts.com quoting Certificate ID: PCS/INT/2026/0090.

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